

AI and ML in Soil Agriculture

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March 28, 2024

1 Challenge to feed 10 billion by 2050

- [?] suggests that this goal is possible but there are many specific challenge areas, such as reducing food waste, and increasing production without increasing land use.
- Aside from reforesting unproductive agricultural areas, we also need to protect and restore peatlands.
- We also need to reduce the damaging affect from pesticide and fertiliser run-off into natural eco-systems.
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2 Soils are crucial for future food production

- Over-watering can deplete soils of nutrients and raise the salt-table.
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3 AI and ML to the rescue?

- Not likely to replace existing methods but maybe complementary
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4 AI for measuring soil health

- Image analysis can pick up variations in crops to enable underlying problems with soil health to be exposed.
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5 Trends for AI in agriculture

- Initial work in solving specific problems - such as killing weeds with lasers.
- Holistic approaches to whole community solutions. Need to make it work for the community - not focus on techno-solutions.
- Google maps parallel - need solution that is equivalent - makes every farmer small percentage more efficient.
- As opposed to more techno solutions - lasering weeds - which may only be viable in limited situations.

6 Points

- Solutions for robotic pickers and crop yield assessment
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